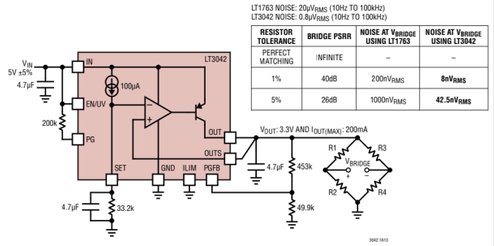
[5] LT3042 - LDO
6V to 5V Analog

TYPICAL APPLICATIONS

Low Noise Wheatstone Bridge Power Supply



LT1763 NOISE: $20\mu V_{RMS}$ (10Hz TO 100kHz)
LT3042 NOISE: $0.8\mu V_{RMS}$ (10Hz TO 100kHz)

RESISTOR TOLERANCE	BRIDGE PSRR	NOISE AT V_{BRIDGE} USING LT1763	NOISE AT V_{BRIDGE} USING LT3042
PERFECT MATCHING	INFINITE	–	–
1%	40dB	$200nV_{RMS}$	**$8nV_{RMS}$**
5%	26dB	$1000nV_{RMS}$	**$42.5nV_{RMS}$**

IN (pins 1-2): LT3042 power input (6V), to be decoupled with CIN.

EN/UV (pin 3): enables/disables the regulator and allows an UVLO threshold (often tied to IN for "always ON").

PG (pin 4): open-collector Power-Good output (requires a pull-up resistor).

ILIM (pin 5): current limit programming/monitoring (often tied to GND if unused).

PGFB (pin 6): input that sets the Power-Good threshold via a resistor divider from VOUT.

SET (pin 7): sets the output voltage via RSET (and CSET reduces noise / provides soft-start).

GND (pin 8): circuit ground reference.

OUTS (pin 9): Kelvin output sense (to be connected to the + of COUT / load point).

OUT (pin 10): regulated output (5V), to be decoupled with COUT.

EPAD (exposed pad): ground + thermal dissipation (to be soldered to the GND plane).

DESIGN NOTE:

Ref Condo for SET PIN (important pour les vibrations P16)
KYOCERA AVX TAJA225K016RNJ (2.2 µF, 16 VDC, condensateur tantale solide SMD moulé, 1206 / 3216 (A case), ±10%, ESR 6.5 Ω).

VOUT ≈ ISET × RSET
ISET ≈ 100 µA

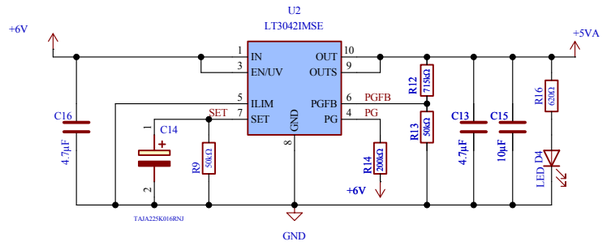
Donc :
RSET ≈ VOUT / ISET
RSET ≈ 5.0 V / 100 µA = 50 kΩ
(=> valeur standard recommandée : 49.9 kΩ)

VOUT(PG_THRESHOLD) ≈ 0.3 V × (1 + RPG2/RPG1)
PGFB_THRESHOLD ≈ 0.3 V

Donc :
RPG2 ≈ RPG1 × (VOUT(PG)/0.3 V − 1)

Exemple (PG OK à ~4.8 V) :
Choix RPG1 = 49.9 kΩ
RPG2 ≈ 49.9 kΩ × (4.8/0.3 − 1) ≈ 748.5 kΩ
(=> valeur standard recommandée : 750 kΩ)

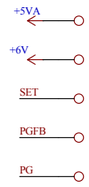
Design guide line



Convertisseur 5VA

About

The LT®3042 is a high performance low dropout linear regulator featuring LTC's ultralow noise and ultrahigh PSRR architecture for powering noise sensitive RF applications. Designed as a precision current reference followed by a high performance voltage buffer, the LT3042 can be easily paralleled to further reduce noise, increase output current and spread heat on the PCB. The device supplies 200mA at a typical 350mV dropout voltage. Operating quiescent current is nominally 2mA and drops to <<1µA in shutdown. The LT3042's wide output voltage range (0V to 15V) while maintaining unitygain operation provides virtually constant output noise, PSRR, bandwidth and load regulation, independent of the programmed output voltage. Additionally, the regulator features programmable current limit, fast start-up capability and programmable power good to indicate output voltage regulation. The LT3042 is stable with a minimum 4.7µF ceramic output capacitor. Built-in protection includes reverse battery protection, reverse current protection, internal current limit with foldback and thermal limit with hysteresis. The LT3042 is available in thermally enhanced 10-Lead MSOP and 3mm × 3mm DFN package



TEST POINT

Comments: LT3042 Analog Devices	Company: OPEN THRUST LAB		Variant:	Git Hash: 349b121
	Board Name: AD7190		Project Name: OPEN_WEIGHT	
Sheet Title: LT3042 - LDO 6V to 5V Analog	File Name: LT3042 - LDO 6V to 5VA.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
Sheet Path: /Project Architecture/LT3042_6V_to_5VA/		Reviewer:	Size: A3	Sheet: 5 of 10